AI-based Concurrent Engineering Facility for future Space Missions design

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Modern space missions call for design environments that are not only technically integrated, but also increasingly adaptive, interactive, and capable of handling complexity from the earliest formulation stages. In this perspective, emerging systems are pushing mission engineering beyond conventional toolchains toward intelligent environments in which requirement interpretation, disciplinary reasoning, and user interaction can be combined within a unified process.

This paper presents the use of the tool STAR (Space mission Twin for Architecture Representation), conceived within the Space It Up! project, as the foundation of a digitalized concurrent engineering environment for space mission design. STAR is a modular digital twin framework conceived to support the definition, analysis, and assessment of space missions through the interaction of interoperable disciplinary modules. The proposed approach interprets its architecture as the virtual counterpart of a Concurrent Engineering Facility (CEF), where the traditional disciplinary contributions are reproduced by specialized modules supported by artificial intelligence and capable of exchanging data, constraints, and design feedback in an iterative way.

Within this framework, a mission can be initiated from requirements expressed in natural language and processed through Artificial Intelligence (AI) - based methods to extract objectives, constraints, and key design drivers. These inputs are propagated across the STAR architecture, where the interaction among the modules progressively structures the mission, identifies the relevant subsystem needs, performs the required simulations and technical assessments, and supports the selection of suitable components and technological solutions needed to realize the overall concept.

Some of STAR's modules incorporate human-machine interaction through Virtual Reality (VR) solutions. In these cases, immersion becomes a means to explore specific mission aspects more effectively, enabling the user to interact with complex scenarios, inspect design implications, and support interpretation and decision-making within the engineering loop. This combination of

module cooperation and interaction extends the role of the digital twin from analysis environment to an active design and operational support entity.

The paper presents an intelligent engineering environment that acts both as a digital counterpart of a CEF and as an enabler of a new user interaction paradigm, in which dedicated interfaces, including VR-based immersive solutions, allow the user to supervise, explore, and influence the mission from a system-level perspective rather than through the traditional discipline-by-discipline approach. The work is developed within Space It Up! (SIU), the Italian national initiative funded by the Italian Space Agency (ASI) and the Ministry of University and Research (MUR).